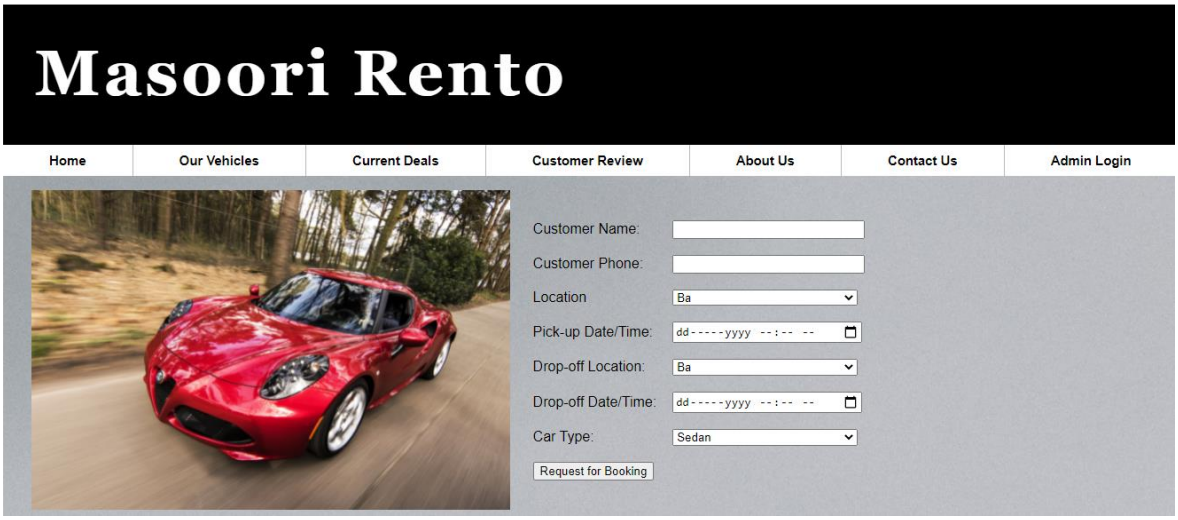
IS221 Semester 1 2023 Finals



1. Write code for txtCusPhone Textbox in asp.net.

   _____

   Solution: <asp:TextBox ID="txtCusPhone" runat="server""">

2. Write code for ddlCarType DropDownList in asp.net with 2 ListItems (Sedan and SUV) in the list.

   Solution:

```
<asp:DropDownList ID="ddlCarType" runat="server" >
        <asp:ListItem>Sedan</asp:ListItem>
        <asp:ListItem>SUV</asp:ListItem>
</asp:DropDownList>
```

3. Drop-off Location option has a dropdown list (ddlDropoffLocation). The following are the options in ddlDropoffLocation: Ba, Lami, Lautoka, Nadi, and Navua. This list uses Locations.xml as a data source. However, Locations.xml is empty. Write XML code that will go inside Locations.xml.

   Solution:

```
<?xml version="1.0" encoding="utf-8" ?>
<locations>
  <location name="Ba" />
  <location name="Lami" />
  <location name="Lautoka" />
  <location name="Nadi" />
  <location name="Navua" />
< locations>
```

4. The following is the structure of RentalCar table in Microsoft Access. What control will be used to connect web form to table in Microsoft Access?

RentalCar

Field Name	Data Type
ID	AutoNumber
Type	Short Text
Pickup_Location	Short Text
Pickup_DateTime	Date/Time
Dropoff_Location	Short Text
Dropoff_DateTime	Date/Time
Cus_Name	Short Text
Cus_Phone	Short Text

_____

Solution: AccessDataSource

5. Write code to extract value from the following:

   ddlCarType: _____

   Solution: ddlCarType.SelectedValue

   txtDropoffDateTime:_____

   Solution: txtDropoffDateTime.Text

6. Customer phone number should only accept numeric values and have 10 digits.
   a. What control should be used to validate txtCusPhone?

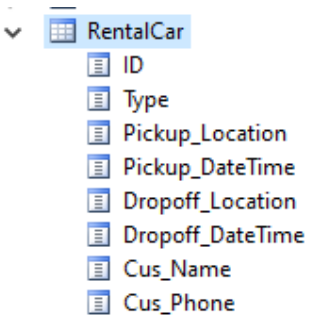
      _____

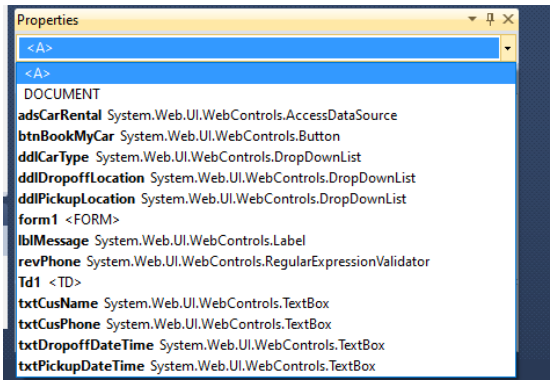
      Solution: RegularExpressionValidator
   b. Write down the validation expression to make sure user does not input any invalid phone number. _____

      Solution: [0-9]{10}
   c. What control should be used to force the customer to enter phone number before submitting the form?

      _____

      Solution: RequiredFieldValidator
7. Upon clicking on "Request for Booking" (btnRequestBooking) button, the user information will be inserted into the database. Write VB code that will handle btnRequestBooking click event.





Solution:

```
Protected Sub btnRequestBooking_Click(sender As Object, e As System.EventArgs) Handles
btnRequestBooking.Click
adsCarRental.InsertParameters("Type").DefaultValue = ddlCarType.SelectedValue
```

```
    adsCarRental.InsertParameters("Pickup_Location").DefaultValue =
ddlPickupLocation.SelectedValue
    adsCarRental.InsertParameters("Pickup_DateTime").DefaultValue = txtPickupDateTime.Text
    adsCarRental.InsertParameters("Dropoff_Location").DefaultValue =
ddlDropoffLocation.SelectedValue
    adsCarRental.InsertParameters("Dropoff_DateTime").DefaultValue = txtDropoffDateTime.Text
    adsCarRental.InsertParameters("Cus_Phone").DefaultValue = txtCusPhone.Text
    adsCarRental.InsertParameters("Cus_Name").DefaultValue = txtCusName.Text
    adsCarRental.Insert()

    lblMessage.Text = "Thank you for choosing us. <br/> We will check the availability and get back
to you soon."
  End Sub
```

8. Masoori Rento has a printer friendly website. It has print.css that will render beautifully when webpage is printed or viewed in print preview mode.
   a. However, there seems something wrong. The following is rendering in browser rather than in print preview mode. How can you fix this?

      _____

      Solution:
      <link href="print.css" rel="stylesheet" type="text/css" media="print"/>
   b. Write CSS to hide everything that is inside header tag.

      _____
      _____
      _____
   c. Solution:
      header{
      display: none;
      }
9. Masoori Rento is your best friend and wants to make the website mobile friendly (540px and below) and approaches you for help. Write media query that can be used to write css for mobile.

   _____
   _____
   _____

   Solution:
   @media only screen and (max-width: 540px) {...}

Name	Email address	Phone Contact		
		Home Phone	Work Phone	Cell Phone
Racheal	abc@company.com	9999-880	9999-881	9999-882

```html
<nav>
  <ul>
    <li><a href="#">Home</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Shop</a></li>
    <li><a href="#">Blog</a></li>
    <li><a href="#">Connect</a></li>
    <li><a href="#">Contact</a></li>
  </ul>
</nav>
```

```css
nav {
  border: 1px solid #333;
  width: 50%;
  margin: 1rem auto;
  background-color: #eee;
}

nav ul {
  list-style-type: none;
  padding: 0;
  margin: 0;
}
nav li {
  padding: 0.5rem 1rem;
  background-color: #f0c808;
}
nav a {
  color: #00bff3;
  text-decoration: none;
  font-size: 1.5rem;
 }
```

Q1

a) One design principle you can follow is to make your design portable. What does it mean by making your design portable?
[2 Marks]

Users with different browsers, operating systems, and computer platform can easily access your Web site.

b) Another design principle is to design for the screen. Your computer monitor, the destination for your Web pages, is different from print-based media. Identify three differences between your computer monitor and print-based media that you must take into account when planning your Web site.
[3 Marks]

- Consider the shape of a computer screen. Paper-based media are portrait oriented, the computer screen is landscape oriented.

- While a piece of paper reflects light, a computer screen has light passing through it from behind. This affects your choices of colors and contrasts.

- Computer screens use much lower resolution than printed page.

Q2

1. Protocol

The first part of the URL "http://" refers to the Internet protocol that is to be used. In the above example the protcol that is used is HTTP.

2. Domain Name

The second part of the URL "www.elearn.usp.ac.fj" refers to server where the document will be stored. The domain is further broken down into a number of parts. The 'www' refers to the World Wide Web. The 'ac' and 'fj' are known as domain suffixes which identifies the type of website it is. In this case, 'ac' means that the server represents an academic institute and 'fj' represents a country (Fiji).

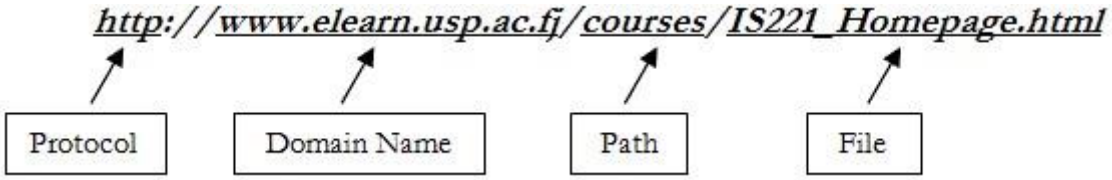
3. Path Name

The third part of the URL "courses/" represents the path name which specifies the directory on the server where the file is stored.

4. File Name

The last part of the URL "IS221_Homepage.html" is the actual file name of the file you are viewing. The filename is in two parts where you have the filename and then the file extension which determines the type of file e.g. .html, .pdf, .doc, etc...

The illustration shows the four parts of an URL.



Q3

List four human factors to consider when building a Web site.
[2 Marks]

- How users will interact with the content
- How users will view the Web pages
- Reading patterns
- Tasks users want to accomplish
- How easily users can navigate the site

- Amount of information to provide
- Accessibility

Q4

- Browser compatibility
- Connection speed differences
- Browser cache and download time
- User computing devices, including monitors
- User operating system

Q5

The rules for the document language so the browser knows how to interpret the HTML code

Q6

Q7

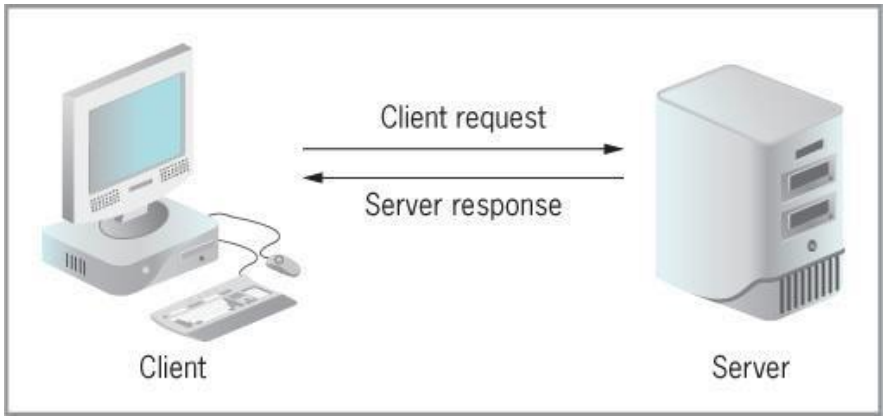
Two-tier system: a system consisting of a client and a server

Client (frontend):

Handles the presentation of the interface to the user

Receives, formats, and presents the results returned from the server

Server(back-end): performs the heavy processing such as calculation

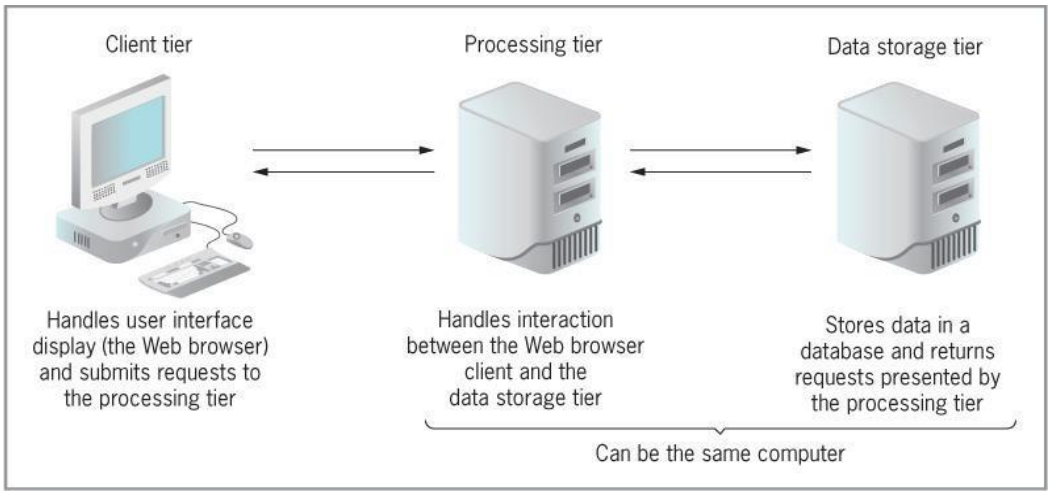


Three-tier (multitier) client/server system: contains three distinct pieces:

Client tier: the Web browser

Processing tier: handles the interaction between the Web browser client and the data storage tier

Data storage tier: stores data in a database and handles requests from the processing tier



Q8

Crafting the look and feel (2 mark)

Grid structure, uniform use of blue and white color scheme and clearly labelled sections etc. afforded a very professional look that is intuitively easy to use.

Design portability (2 mark)

Coding to standards and the use of different stylesheets etc. enable Facebook's site to be accessible on any platform. They also have a mobile version of the main site.

Easy access to your information (2 mark)

Facebook used a great information design that is based on what the users are expected to be able to do and as such structured the main links accordingly hence maintaining the 3-click rule of accessing information.

Planning for Clear presentation of information (2 mark)

Use of the grid structure once again and active use of white space that partitions the various sections of FB with clear headings and icons that are representative of the various activities e.g. mail icon for messaging, etc.

Design for low bandwidth (2 mark)

Minimalistic use of graphics and strategic reusing of universally understood icons.

Q9

Hypertext Transfer Protocol (HTTP) to serve HTML files to web browser clients. Unless your company or organization has a web server and hosts its own content, you must use the services of a web hosting provider. After you choose a server to host your files, you need to select file transfer software and upload the web site files from your development machine to the web server.

Q10

- Add Viewport to head section of pages
- Determine breakpoints which should be driven by content of the web pages
- @ media queries to specify the breakpoints for the mobile phone screen size
- Create stylesheets for the different breakpoints
- Use flexible/relative measurements in CSS
- Stack and remove columns
- Minimise choices yet maintain functionalities
  Use drop down menus, slide shows etc.

Q11

Requirements specification
Activities

- Determine how the customer wants to use the site e.g. will there be credit card payments etc. Will there be a catalogue for customers to view and select the kava packs they want. How about delivery, will you customers need to provide their delivery address etc.
- Type of website in this case will be an e-commerce website therefore think of common elements that an e-commerce website should have.
  Since it's going to be an e-commerce website you will need to think of an appropriate domain name e.g. renting online a .com domain would be most appropriate.
- Confirm requirements and specifications with client.

Artefacts
• Registered domain name (www.uniroomsforrent.com.fj)
• Requirements specifications document which will consist of: o Technical specifications e.g. online searchable catalogue, form to enter customer's details etc and a database to store catalogue details and if need be customers' details.
o Design requirements e.g. colour scheme or a theme to be used for the website that is in line with the Uni rooms for rent concept
o Page layout sketches e.g. a sketch of what the site may look like o Audience definition e.g. who your potential customers are, will they be just local customers or international customers
Project/website specifications report

Information design and taxonomy creation
` **Activities**
• Sorting out the web pages the customer wants for instance what content to have on what page and how to structure content and hyperlinks etc. Think in terms of the layouts discussed in readings, customer might want to have a featured product/rental specials or something
• Sketch of what the pages and the website may look like and navigation between pages etc.
Artefacts
Information design
• A sketch of the various web pages and how they should be interlinked based on their content